Multispectral Remote Sensing-Driven Evaluation of Chlorophyll in Tea Plant Canopies

Jiaxing Xie, Liye Chen, Jiatao Wu, Zonghong Li, Yazhong Chen, Meiyi Lu, Yingxin Zou, Peng Gao, Zheng Shen, Daozong Sun, Weixing Wang, and Jun Li

***Abstract*—This article introduces an uncrewed aerial vehicle (UAV)-based multispectral model, YOLO-SPAD, for rapid, nondestructive estimation of the relative chlorophyll content of tea leaves. This is crucial for predicting growth conditions, implementing precise irrigation and fertilization, and increasing tea yields. The model leverages the spectral separation of the tea tree canopy and the YOLOv8 neural network architecture for image segmentation. Multispectral imagery of the standardized tea plantation was captured by a UAV equipped with a five-channel camera, while corresponding soil plant analysis development (SPAD) values were measured using a SPAD-502Plus device. Based on the YOLOv8 model, it is proposed to incorporate a spectral fusion (SF) module to better adapt to multispectral characteristics. Four deep learning models for image segmentation were further compared. The improved YOLOv8 model achieved a segmentation mIoU of 91.61% for the tea tree canopy, outperforming the YOLOv5 (90.65%) and DeepLabv3+ (86.14%), which were improved using the same method. Five different band combinations, 43 vegetation indexes, and 40 texture features were analyzed to construct mapping transformation pairs for reflectance in the segmented area. This was applied to the SPAD prediction head to implement YOLO-SPAD. The YOLO-SPAD model was used to predict tea canopy SPAD using UAV multispectral imagery with a coefficient of determination (R^2) of 0.85 and a low error (root mean square error (RMSE) =2.14, mean absolute error (MAE) =1.79), providing accurate and stable predictions. This model supports dynamic monitoring of tea tree growth via UAV remote sensing, aiding crop nutrition improvement and precision agriculture implementation.**

***Index Terms*—Mapping transformation, soil plant analysis development (SPAD) prediction, spectral fusion (SF), uncrewed aerial vehicle (UAV)-based multispectral, YOLOv8.**

I. Introduction

TEA, as a widely consumed beverage with significant cultural and economic importance [1], is derived from tea leaves. Furthermore, previous studies [2] have demonstrated that the chlorophyll content of tea leaves is strongly associated with their growth and quality. Therefore, measuring the chlorophyll level during the growth of tea trees can predict the health and development of tea trees, and may provide a reference for the growth status of tea trees and the suitability of the environment. Traditional methods for measuring chlorophyll content involve either extracting chlorophyll from leaf samples [3] or employing optical techniques [4]. However, these approaches are typically confined to standard laboratories, are time-consuming, and can harm crop leaves, rendering them unsuitable for large-scale implementation in agricultural production.

In the 1980s, Japan's Minolta Company developed a dual-wavelength absorbance-based chlorophyll meter known as soil plant analysis development (SPAD) (later improved as the SPAD-502). This device garnered significant attention within the scientific community [5]. Existing research [6] demonstrates that SPAD is closely linked to the relative chlorophyll and nitrogen content in leaves, exhibiting a strong correlation with the chlorophyll concentration in plant leaves. Essentially, SPAD directly reflects the growth status of crops: the higher the SPAD value, the better the growth status [7], [8]. In practical applications, researchers have harnessed SPAD for various purposes. For instance, Pallavolu et al. [9] utilized SPAD to establish the relationship between leaf chlorophyll content (LCC) and seed yield. Duan et al. [10] collected SPAD data at different growth stages of rice, analyzing the connection between crop leaf area index (LAI) and SPAD. Their work contributed to improved accuracy in estimating rice yield. Wan et al. [11] discovered that SPAD holds great potential for converting into LCC and leaf greenness. Consequently, measuring and estimating SPAD values can

Received 15 January 2025; revised 7 May 2025, 23 June 2025, and 21 August 2025; accepted 2 September 2025. Date of publication 8 September 2025; date of current version 16 September 2025. This work was supported in part by the Key-Area Research and Development Program of Guangdong Province under Grant 2023B0202090001; in part by the China Agriculture Research System of MOF and MARA, China, under Grant CARS-32-10; in part by Guangdong Science and Technology Innovation Cultivation Special Fund Project for College Students ("Climbing Program" Special Fund), China, under Grant pdjh2023a0074; in part by the Ministry of Education Industry Education Collaborative Education Project under Grant 230702595161723; and in part by the College Student Innovation and Entrepreneurship Training Program Project under Grant 2023105641146. *(Corresponding authors: Weixing Wang; Jun Li.)*

Jiaxing Xie and Daozong Sun are with the College of Electronic Engineering (College of Artificial Intelligence), South China Agricultural University, Guangzhou 510642, China, and also with the Engineering Research Center for Monitoring Agricultural Information of Guangdong, Guangzhou 510642, China (e-mail: xjx1998@scau.edu.cn; sundaozong@scau.edu.cn).

Liye Chen, Jiatao Wu, Zonghong Li, Yazhong Chen, Meiyi Lu, Yingxin Zou, and Zheng Shen are with the College of Electronic Engineering (College of Artificial Intelligence), South China Agricultural University, Guangzhou 510642, China (e-mail: scau-chenliye@foxmail.com; wjt_@stu.scau.edu.com; 13423567789@163.com; chenyazhong0@gmail.com; meiyilu640@gmail.com; 1379815078@qq.com; 2032772361@qq.com).

Peng Gao is with the College of Electronic Engineering (College of Artificial Intelligence), South China Agricultural University, Guangzhou 510642, China, and also with the Engineering Research Center for Monitoring Agricultural Information of Guangdong, Guangzhou 510642, China (e-mail: gaopeng.peng@scau.edu.cn).

Weixing Wang is with Zhujiang College, South China Agricultural University, Guangzhou 510900, China (e-mail: weixing@scau.edu.cn).

Jun Li is with the College of Engineering, South China Agricultural University, Guangzhou 510642, China (e-mail: autojunli@scau.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3607512

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serve as an effective method for predicting crop growth conditions.

With the advent of uncrewed aerial vehicle (UAV)-based plant protection and agricultural remote sensing, researchers have increasingly turned to UAV imagery for studying crop conditions [12]. Plant phenotyping research has also gained prominence. Low-altitude UAV remote sensing platforms can be used to estimate crop phenotypic information and biomass parameters because of their high flexibility, low cost, and ease of operation, and are expected to become a tool for agricultural remote sensing [8], [13], [14], [15]. Multispectral and hyperspectral cameras are commonly employed in academic circles to collect crop vegetation reflectance data across different spectra [16]. For instance, Xu et al. [17] utilized a UAV equipped with a hyperspectral sensor to capture hyperspectral reflectance information from ground crops. They established a vegetation index and corresponding texture model, successfully estimating rice aboveground biomass (AGB). Jiang et al. [18] combined UAV remote sensing technology with various machine learning algorithms, including random forest (RF), efficient neural network (Enet), artificial neural network (ANN), and partial least squares regression (PLSR). Their work focused on predicting the biological phenotypic parameters of wheat during the jointing and booting stages. In the study by Xie et al. [19], a UAV equipped with a multispectral camera captured multispectral remote sensing images of lychee canopies. They extracted spectral reflectance, vegetation indexes, and texture features. By employing support vector regression (SVR), RF, K-nearest neighbor regression (KNR), and ridge regression (RR) models, they conducted an inversion estimation study on the SPAD value of lychee canopies.

However, the aforementioned research primarily focuses on employing mathematical modeling and machine learning techniques to establish correlations between biological parameter indicators and UAV remote sensing image data, as well as the relationship between crop SPAD values and unique information extracted from multispectral images. These topics have been extensively discussed and researched. In practical usage, selecting the region of interest (ROI) within an image often involves manual intervention. Researchers manually mark the ROI, and the estimation outcome of the SPAD value is significantly influenced by human subjective factors. With the popularization of automated smart agricultural production methods in the future, it is necessary to conduct more in-depth research and discussion on the application value and actual effect of multispectral images in predicting crop biological parameters [20].

In recent years, the scientific community has witnessed a proliferation of deep neural networks (DNNs), fueled by advancements in computer hardware and artificial intelligence [21]. These DNN-based methods find diverse applications in UAV remote sensing, particularly within the agricultural domain. For instance, Yang et al. [22] combined multispectral imaging with deep learning models to detect defects in potatoes. Their approach significantly enhanced the utilization of multispectral image information. Blekanov et al. [23] annotated farmland UAV multispectral source images with varying nitrogen levels. Leveraging the U-Net architecture and an improved semantic segmentation model, they directly estimated the nitrogen status of plants, achieving effective classification.

The purpose of this study is to explore the possibility of using UAV multispectral imagery for SPAD value estimation and monitoring in view of the lack of practice in crop biomass prediction using UAV remote sensing technology and deep learning. This may promote the implementation of agricultural condition monitoring and crop growth status prediction, and realize precision agriculture.

Therefore, this study attempts to establish a tea tree SPAD estimation model based on deep learning and drone multispectral images—YOLO-SPAD. Using deep learning algorithms, the tea tree canopy in the planting area is segmented, and the segmented Mask is used as the ROI; the spectral reflectance in the region of most interest is extracted, the vegetation index and texture features are calculated, and the effective vegetation index and texture features are determined by analysis and verification; the mask-feature map transformation pairs are generated using reasonable vegetation index and texture features, and the RF model is integrated to construct the tea tree canopy SPAD prediction model and evaluate it.

II. Materials and Methods

A. Study Area

The data collection took place at the tea tree nursery of South China Agricultural University, located in Guangzhou City, Guangdong Province (coordinates: 113°36'E, 23°15'N), as depicted in Fig. 1. According to the Köppen system of climate classification, the area has a subtropical humid hot summer climate [24], with an average annual temperature of 21 °C and an average annual total precipitation of 1682 mm. Within the nursery, tea trees of the "Huanong No. 1" variety, aged between 3 and 4 years, are cultivated.

In this experiment, the tea trees in the planting area were divided into five nonrepetitive parts, and different soil moisture content treatments were performed, namely, 85%–95%, 75%–85%, 55%–65%, and 45%–55% of the field water holding capacity and natural state. Each part was used to collect training data samples. The soil moisture level was determined by a JK-300F soil moisture meter (JIEKESI, Xinghua, Jiangsu, China). The purpose of this is to make the tea trees in the planting area have different chlorophyll contents under different water stress.

B. Research Process Flow

The research process for predicting tea tree SPAD values using UAV multispectral images is depicted in Fig. 2.

Initially, a UAV was precisely controlled to capture multispectral images of the tea tree canopy, forming an extensive image dataset. Concurrently, a SPAD-502Plus chlorophyll meter was deployed on the ground to simultaneously collect SPAD values directly from the tea tree canopy. Following the accurate annotation of the tea tree canopy within the multispectral images, model training was conducted. The

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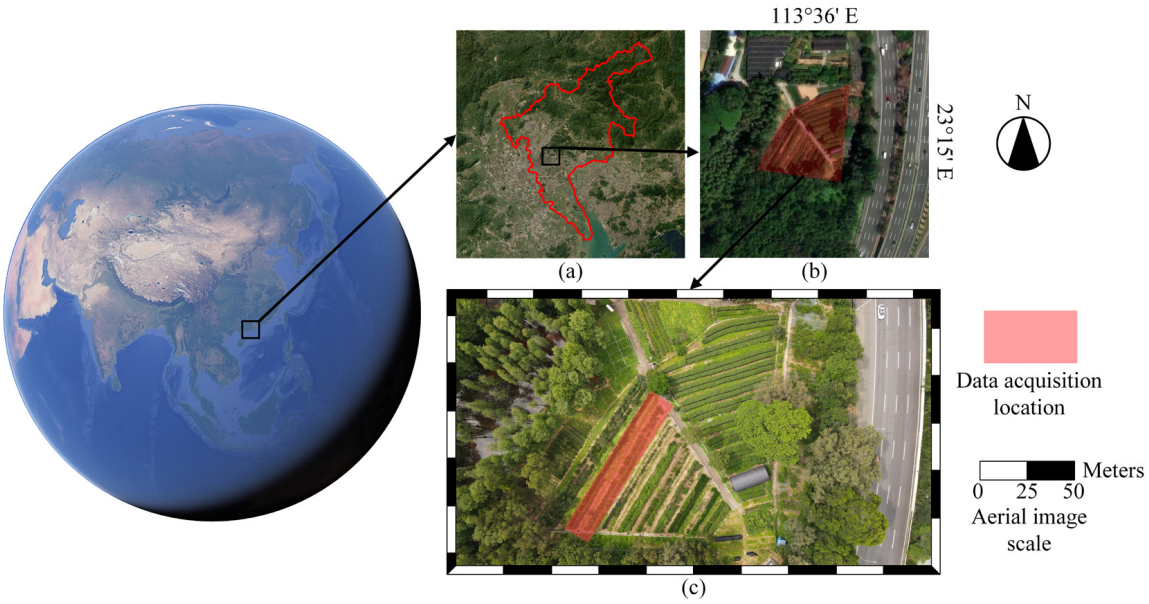


Fig. 1. Study area. (a) Geographical location of Guangzhou in China. (b) South China Agricultural University tea tree plantation location. (c) Aerial view of the research area.

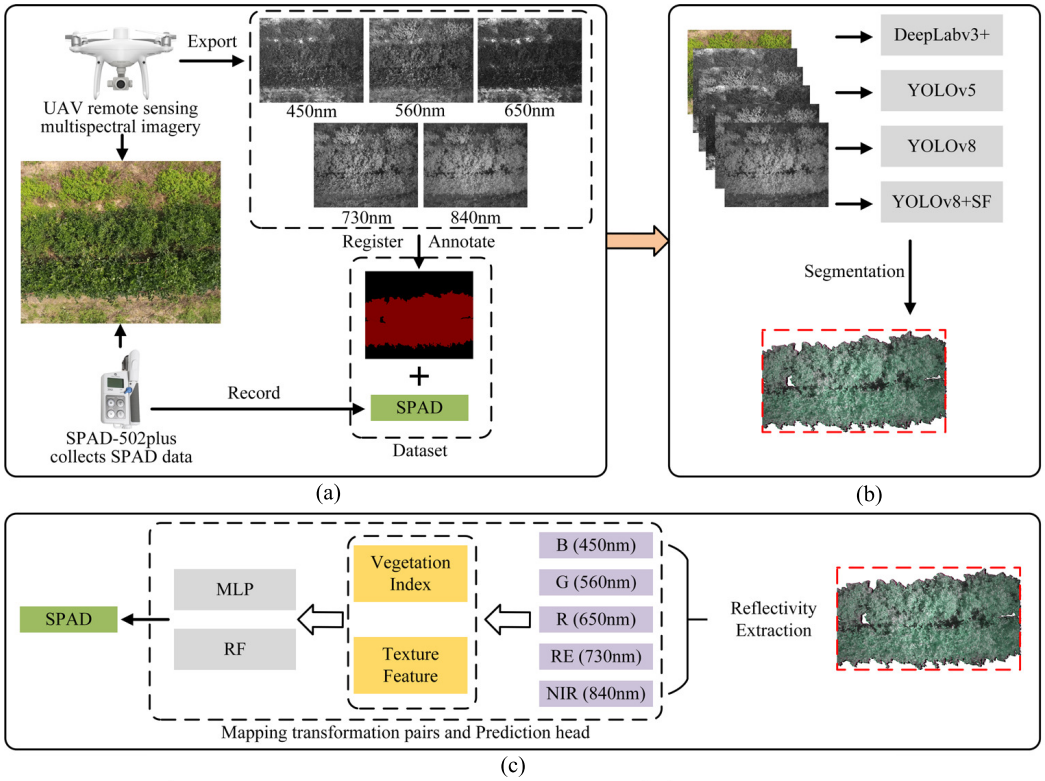


Fig. 2. Research process flow. (a) Acquire data. (b) ROI segmentation and model fusion. (c) Prediction-head of the YOLO-SPAD and Chlorophyll content estimation.

subsequent step involved a comparative analysis of the results obtained from different models for tea tree canopy segmentation. In addition, by screening different multispectral vegetation indexes and texture features, mapping transformation pairs were constructed. These pairs were instrumental in forming a SPAD prediction head, which contributed to enhancing the YOLOv8 model. Through the proposed YOLO-SPAD model, the SPAD value of the tea tree canopy can be measured using drone multispectral data, and the model effect can be analyzed.

C. Data Acquirement

This study used the DJI Phantom 4 Multispectral Edition (DJI, Shenzhen, Guangdong, China) to capture aerial multi-

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TABLE I
UAV AND MULTISPECTRAL CAMERA TECHNICAL PARAMETERS

DJI Phantom 4 multispectral version	
Flight duration	about 20 minutes
Relative working height	10~100m
Flight preparation time	5-10 minutes
Takeoff and landing area	$5\times5m^2$
Aircraft power	electric drive
Ground sampling distance (GSD)	(H/18.9) cm/pixel
Working temperature	0-40°C
Maximum withstand wind speed	10m/s
Image sensor	Equipped with 6 1/2.9-inch CMOS multispectral cameras Effective pixels of a single sensor are 2.08 million
Spectrum and their resolution	blue (B): 450nm±16nm green (G): 560nm±16nm red (R): 650n±16nm red-edge (RE): 730nm±16nm near-infrared (NIR): 840nm±26nm
Field of view (FOV)	62.7°
Focal length	5.74mm (35mm format equivalent: 40mm)
Aperture	f/2.2
Maximum image size	1600×1300

spectral images of the tea tree canopy in the study area. The multispectral camera on board has the ability to capture images of visible light and five bands: blue, green, red, red-edge, and infrared. The key technical characteristics of the DJI Phantom 4 drone's multispectral version are summarized in Table I.

In this study, the multispectral image dataset was collected from March to September 2023, with 4168 images. Among them, there are 1224 images of the tea tree's spring shoot growth period (March to May), 1196 images of the summer shoot growth period (June to July), and 1766 images of the autumn shoot growth period (July to September), covering the entire life cycle of tea leaves from budding to senescence. The data collection strategy was selected during clear daytime. After determining suitable take-off and landing points within the tea nursery plantation, drone attitude correction and multispectral camera calibration were performed. The flight plan was created through the DJI GS Pro Ground Station Professional Edition App on a mobile device. Then, the drone was manually controlled to take off and fly to hover above the mark marked on the ground to collect multispectral images, so that the captured image range can roughly cover the marked area. Fig. 3 displays a series of composite multispectral images of tea trees captured by a UAV. At the same time, the SPAD value of tea leaves was collected synchronously on the ground through the SPAD-502Plus chlorophyll meter.

Tea tree canopy SPAD data was collected and measured using the SPAD-502Plus chlorophyll meter. This instrument assesses the relative absorbance of 650- and 940-nm wavelength light after it passes through the leaves, allowing us to calculate the relative chlorophyll content in the leaves. To collect data, five sampling points were selected in each area under different water stress treatments. These points were distributed in the center and four corners of the area, i.e., according to the five-point sampling method. During each SPAD measurement, three tea leaves were randomly chosen within the sampling point. The instrument was positioned at the center of the leaves, avoiding the leaf veins. The average SPAD value of the three tea trees represented the final SPAD value for that specific sample point.

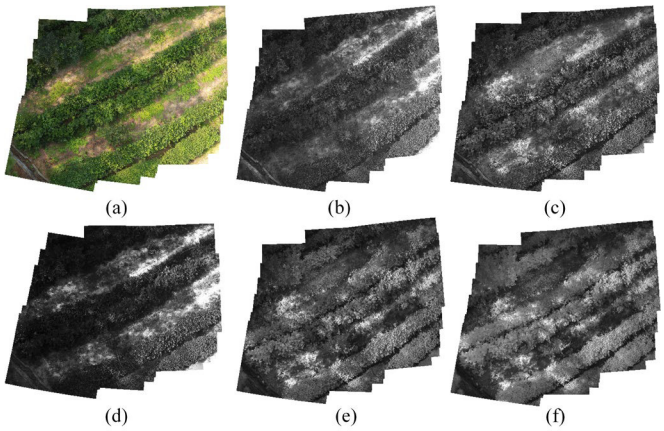


Fig. 3. Set of multispectral photographs taken by a UAV. (a) RGB. (b) 450 nm (blue). (c) 560 nm (green). (d) 650 nm (red). (e) 730 nm (red-edge). (f) 840 nm (near-infrared).

D. Multispectral Image Registration

This article uses the SimpleITK library to register multispectral images taken by drones [25]. The mean square error

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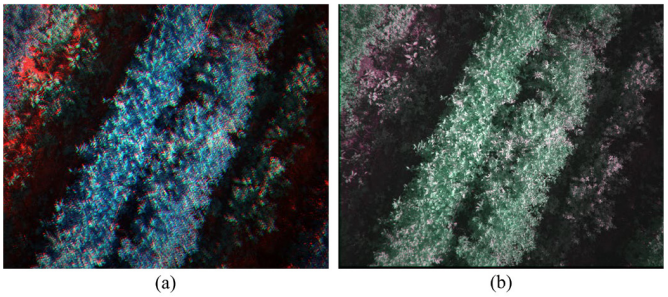


Fig. 4. Multispectral images (a) before and (b) after registration.

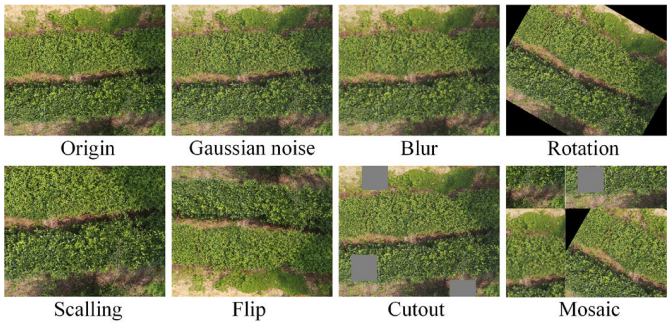


Fig. 5. Original image and data-enhanced image.

metric function is used to evaluate the similarity between multispectral images. The regularized gradient descent optimizer, translation transformation, and linear interpolation are configured to align the multispectral images with the reference registration. Fig. 4 shows a pair of unregistered images and the registered images.

E. Data Preprocessing and Image Enhancement

The training of deep learning networks necessitates a substantial amount of annotated data for feature extraction and learning [26]. However, noise and errors within the data can significantly impact model fitting and overall performance. When dealing with a large number of mislabeled or inaccurate images, the model's ability to learn effectively and extract correct features may be compromised. Conversely, if the dataset is limited, the network might overfit to the available data, inadvertently learning unexpected noise and interference information from the images. This can adversely affect the network's calculation accuracy. To this end, this article proposes a tea tree canopy multispectral image data enhancement pipeline, which introduces the use of Gaussian noise, blur, random rotation, random scaling, vertical or horizontal flipping, adding occlusion, etc. in the program, and then a part of it is spliced by random scaling, random cropping, and random arrangement through Mosaic data enhancement, which enriches the image dataset and enhances the robustness of the deep learning network. It should be emphasized that the parameters of the data augmentation methods used in this article are all randomized by the program. Fig. 5 provides an illustrative example, showcasing both the original image and its enhanced counterpart.

TABLE II
DATASET USED IN THIS STUDY CONTAINS THE NUMBER

Dataset	Sample size		Total samples
RGB and multispectral images	Training	2470	4168
	Validation	834	
	Test	864	
SPAD	Modeling	102	144
	Test	42	

In order to test the prediction results of tea tree canopy segmentation and extract reflectivity, and further test the prediction model capability and effect of tea tree canopy SPAD, a total of 864 RGB and multispectral images taken on March 7, April 17, July 21, August 5, and September 21, 2023, and 144 SPAD sample data collected were retained to construct UAV tea tree multispectral images and SPAD datasets to test the tea tree canopy segmentation model performance and establish the tea tree canopy SPAD prediction head. Table II shows the data and sample size used in this study.

F. YOLO-SPAD

The YOLO-SPAD model, proposed in this article for predicting tea tree canopy SPAD from UAV multispectral images, consists of two main components: a fully convolutional segmentation model and a SPAD prediction head based on YOLOv8 (as depicted in Fig. 6). To elaborate, the process begins by reducing the size of the high-resolution spectral images. Subsequently, a segmentation model is employed to determine the position and boundaries of the tea tree canopy. The SPAD prediction head then maps the spectral reflectance of the tea tree canopy to vegetation indexes and texture features, leveraging the segmentation results. This specific feature mapping transformation enhances the model's effectiveness. Finally, ensemble learning is introduced to complete the prediction of the SPAD of the tea tree canopy.

1) YOLOv8: This study used the YOLOv8 segmentation algorithm as the basic model for tea tree canopy segmentation. Within the field of computer vision, the academic community has closely scrutinized the YOLO series of algorithms [27]. Notably, YOLOv8 stands out due to its reduced model size while maintaining high accuracy. It can be effectively trained on a single GPU or deployed on more cost-efficient devices. The YOLOv8, proposed by Ultralytics, serves multiple purposes, including target recognition, classification, and segmentation. When compared to YOLOv5, YOLOv8 introduces several key modifications: The C2f module replaces the C3 module. The first 6×6 Conv in the backbone network is replaced with 3×3 Conv. Two 1×1 Conv convolutions are removed. The first 1×1 Conv in the Bottleneck is replaced with 3×3 Conv. The detection head employs a pair of separate structures. These enhancements collectively contribute to a more robust overall model.

The YOLOv8 model is a deep learning-based instance segmentation model. Unlike semantic segmentation models, the fundamental concept of instance segmentation involves dividing the image into a grid system. Each grid's category

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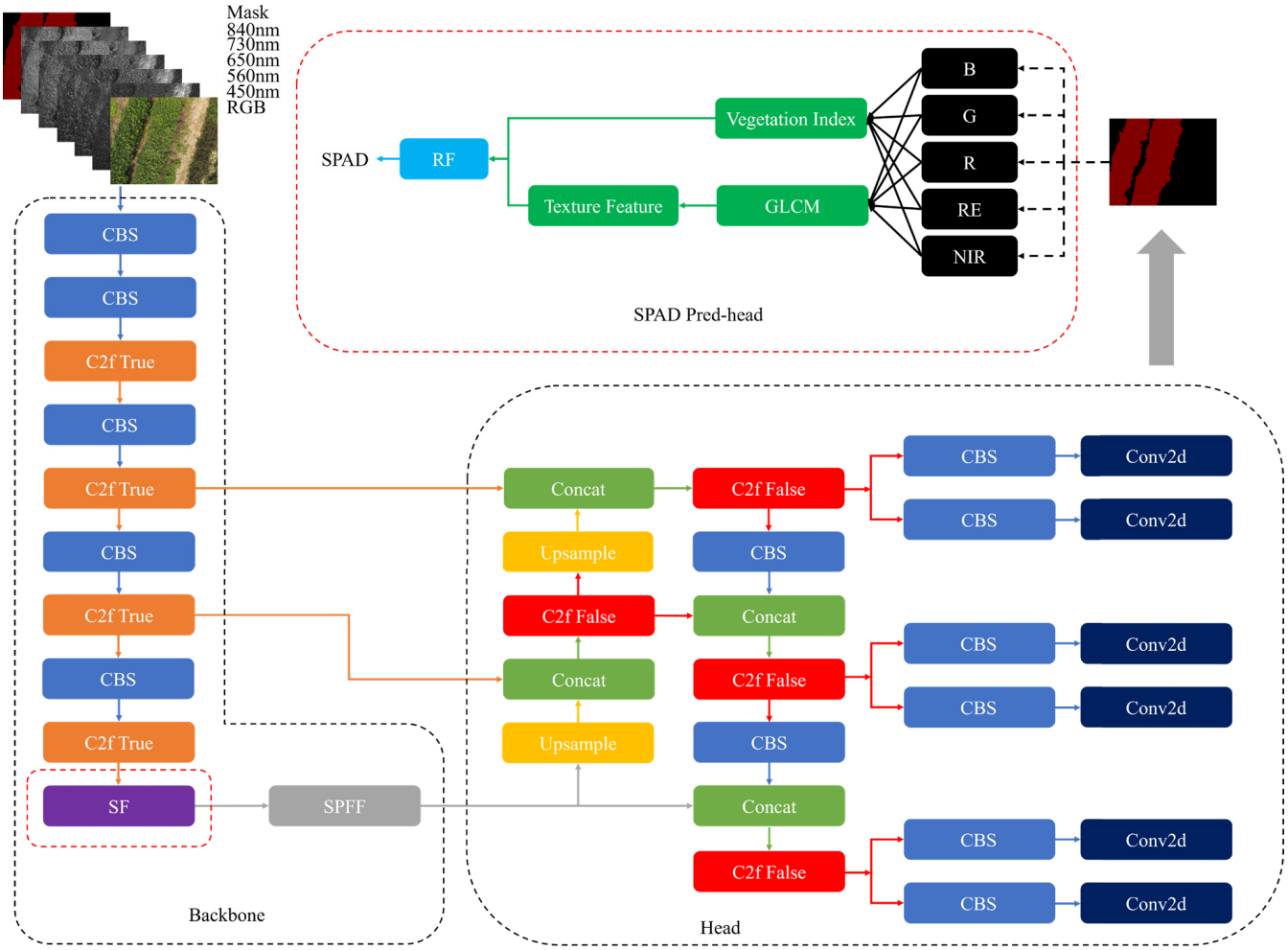


Fig. 6. YOLO-SPAD network structure.

and bounding box are predicted, and the image's pixels are then assigned to a specific instance object based on the bounding box's position and size information. In essence, the YOLOv8 segmentation model operates in two stages. While it may sacrifice some speed in segmentation, it compensates by delivering higher accuracy and precision, and it exhibits superior multiscale characteristics.

In this research, given the unique nature of UAV aerial photography, the segmentation outcomes of the tea tree canopy are vital for the ensuing inversion and prediction of SPAD values. Therefore, in the study of the SPAD prediction model for tea tree canopy, it should be ensured that the ROI is selected correctly, that is, the accuracy of the segmentation results of the deep learning neural network on UAV multispectral images.

2) Spectral Fusion: The fusion of multispectral image features is the primary objective to enable the model to accomplish multispectral segmentation. The feature maps extracted by DNNs contain rich information that ensures a comprehensive understanding of the model's input data. However, this can consume a significant amount of computational resources and increase the model's complexity. The spectral fusion (SF) module integrates dilated convolution and multiscale sampling modules to generate fused feature maps through a series of convolution transformations. This process synthesizes multiband fusion features from original features at a lower cost, thereby reducing the high computational complexity associated with extracting all band features. As a result, it improves the accuracy of tea tree canopy segmentation after SF. The structure of the SF module is depicted in Fig. 7. The proposed SF-enhanced YOLOv8 model is based on the module's ability to adaptively select and adjust the weight of each band feature map, effectively capturing and representing key spectral features of the tea tree canopy.

3) Vegetation Index and Texture Feature Mapping: The segmentation outcome mask of the model is utilized as the ROI to extract the reflectance of the canopy from each spectral image. The reflectance, represented as R_i, is computed in accordance with the following equation:

$$R_i = \frac{D_i}{D_{\text{bn}}} R_{\text{bn}} \tag{1}$$

in the formula, R_i represents the reflectance of the target. D_i represents the quantitative value of the target. R_{bn} represents the reflectance of the calibration plate. D_{bn} represents the quantitative value of the calibration plate.

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TABLE III
VEGETATION INDEX FOR SPAD PREDICTION STUDIES

Vegetation index	Formula	Literature
CIgreen	CIgreen=(NIR/G)-1	[28]
CIre	CIre=(NIR/RE)-1	[29]
CVI	CVI=(NIR/G)(R/G)	[30]
DVI	DVI=NIR-R	[31]
EVI1	EVI1=2.5×((NIR-R)/(NIR+(G×R)-(G×B)+0.1))	[32]
EVI2	EVI2=2.5×(NIR-R)/(NIR+2.4×R+1)	[32]
MCARI	MCARI=((RE-R)-0.2×(RE-G))×(RE/R)	[33]
MNVI	MNVI=1.5×(NIR^2-R)/(NIR^2+R+0.5)	[34]
MSAVI	MSAVI=0.5×(2×(NIR+1)-sqrt($(2\times NIR+1)^2$-8×(NIR-R)))	[33]
NLI	NLI=(NIR^2-R)/(NIR^2+R)	[34]
NDVI	NDVI=(NIR-R)/(NIR+R)	[35]
OSAVI	OSAVI=1.16×(NIR-R)/(NIR+R+0.16)	[35]
RDVI	RDVI=(NIR-R)/sqrt(NIR+R)	[36]
RVI1	RVI1=NIR/R	[37]
RVI2	RVI2=NIR/G	[37]
SIPI	SIPI=(NIR-B)/(NIR+B)	[38]
TCARI	TCARI=3×(RE-R)-0.2×(RE-G)(RE/R)	[39]
TVI	TVI=sqrt(120×(NIR-R)/(NIR+R)+0.5)	[40]
MTCI	MTCI=(NIR-RE)/(RE-R)	[41]
VDVI	VDVI=(2×G-R-B)/(2×G+R+B)	[42]
ENDVI	ENDVI=(RE+G-2×B)/(RE+G+2×B)	[29]
IPVI	IPVI=NIR/(NIR+R)	[43]
NRI	NRI=R/(RE+NIR+R)	[44]
TNDVI1	TNDVI1=sqrt((NIR-R)/(NIR+R+0.5))	[45]
TNDVI2	TNDVI2=sqrt((NIR-R)/(NIR+R)+0.5)	[45]
MDD	MDD=(NIR-RE)-(NIR-R)	[46]
NDRE	NDRE=(NIR-RE)/(NIR+RE)	[47]
RECI	RECI=(NIR/RE)-1	[29]
GSAVI	GSAVI=1.5×(NIR-G)/(NIR+G+0.5)	[46]
DATT	DATT=(NIR-RE)/(NIR-R)	[48]
NREI	NREI=RE/(RE+NIR+G)	[49]
GRVI	GRVI=NIR/B	[50]
MEVI	MEVI=2.5×(NIR-RE)/(NIR+6×RE-7.5×G+1)	[49]
NPCI	NPCI=(R-B)/(R+B)	[51]
MRGBVI	MRGBVI=(RE+2×G-2×B)/(RE+2×G+2×B)	[52]
ENGBVI	ENGBVI=(RE×NIR+2×G-2×B)/(RE×NIR+2×G+2×B)	[52]
MRNGBVI	MRNGBVI=(RE×NIR+G-B)/(RE×NIR+G+B)	[52]
GNDVI	GNDVI=(NIR-G)/(NIR+G)	[28]
SAVI	SAVI=1.5×(NIR-R)/(NIR+R+0.5)	[53]
SR	SR=NIR/RE	[54]
MTVI	MTVI=1.5×(1.2×(RE-G)-2.1×(R-G)	[55]
REV	REV=(NIR-RE)/(NIR+RE)	[47]
MSR	MSR=(NIR/R-1)×(NIR/R+1)	[56]

Owing to the influence of leaf structure, canopy characteristics, and background, spectral reflectance has a limited impact on the model's performance [55], [57], [58], [59], [60], [61]. Therefore, in the realm of agricultural remote sensing research, the vegetation index and texture characteristics serve as key independent variables for the inversion and prediction of crop biomass parameters. Drawing from existing research and literature, this study identified 43 vegetation indexes used to estimate the SPAD value of the tea tree canopy. These include the ratio vegetation index (RVI), enhanced vegetation index (EVI), and normalized difference red edge (NDRE) index. For further details, please refer to Table III.

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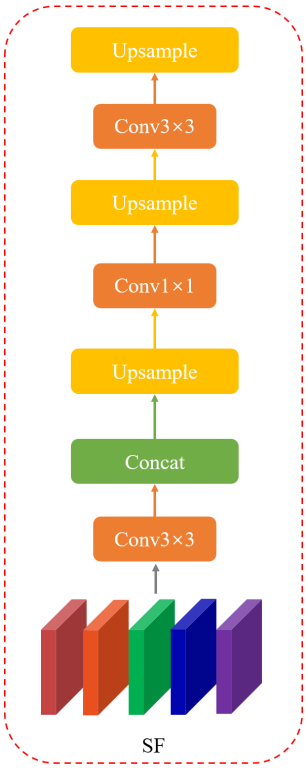


Fig. 7. SF structure.

Texture features are calculated using the gray-level co-occurrence matrix (GLCM), which is a reflection of the internal characteristics of plants on remote sensing images [62]. The extraction from each wavelength spectral image encompasses the mean, variance, homogeneity, contrast, dissimilarity, entropy, angular second moment (ASM), and correlation. The calculations are as follows:

$$\text{Mean} = \sum_i \sum_j i \times p(i,j) \tag{2}$$

$$\text{Variance} = \sum_i \sum_j (i-u)^2 p(i,j) \tag{3}$$

$$\text{Homogeneity} = \sum_i \sum_j \frac{1}{1+(i,j)^2} p(i,j) \tag{4}$$

$$\text{Contrast} = \sum_{n=0}^{N_g} n^2 \left\{ \sum_{i=1}^{N_g} \sum_{j=1}^{N_g} p(i,j) \atop |i-j| = n \right. \tag{5}$$

$$\text{Dissimilarity} = \sum_{n=1}^{N_g-1} n \left\{ \sum_{i=1}^{N_g} \sum_{j=1}^{N_g} p(i,j) \atop |i-j| = n \right. \tag{6}$$

$$\text{Entropy} = -\sum_i \sum_j p(i,j) \log(p(i,j)) \tag{7}$$

$$\text{ASM} = \sum_i \sum_j \{p(i,j)\}^2 \tag{8}$$

$$\text{Correlation} = \frac{\sum_i \sum_j (i,j)\, p(i,j) - \mu_x \mu_y}{\sigma_x \sigma_y}. \tag{9}$$

Among them, $p(i, j)$ represents the (i, j)th value in the GLCM. N_g denotes the number of distinct gray levels in the image. μ_x and σ_x are the mean and standard deviation (SD) of the x row in the co-occurrence matrix, respectively. Similarly, μ_y and σ_y represent the mean and SD of the y column in the co-occurrence matrix, respectively.

This research employs the Pearson correlation coefficient to assess correlations, selecting the optimal vegetation indexes and texture features for model development by comparing these correlations. A linear regression analysis was conducted on the multispectral vegetation indexes and texture features (independent variables) and the SPAD values (dependent variables) within the experimental area. Step-wise regression (SWR) is applied to the vegetation indexes and texture features that exhibit a higher R^2 value [63], [64], [65], thereby retaining the most effective factor combination for predicting SPAD values [66].

4) SPAD Prediction Head: The SPAD prediction head serves as a crucial component of the overall YOLO-SPAD model, facilitating the SPAD prediction of the tea tree canopy. In the realm of crop biomass estimation and plant phenotype research, it has been confirmed that the optimal method for SPAD estimation involves the use of fused multimodel and ensemble learning [67]. The image target detection and segmentation algorithm based on deep learning is mainly used for object detection and classification. Its output is the target anchor box or pixel category label, which can achieve good results in the discrimination of image target categories. However, the crux of crop canopy SPAD prediction lies in regression. How to fit a better image fusion regression model is the focus of this study. In addressing this issue, it has been observed that the effects and time overhead obtained by ANNs are generally inferior to those achieved by traditional machine learning methods [68].

RF is an ensemble learning methodology that employs multiple decision trees for the training and prediction of samples. It operates by randomly selecting a subset of features and training data. The prediction label that occurs most frequently for the same prediction data is considered as the final prediction outcome. This model is particularly adept at handling high-dimensional data in remote sensing images [69], [70], [71].

Therefore, based on the segmentation model, an RF-based SPAD prediction head is designed. This method is implemented by mapping the transformation of vegetation spectral reflectance. The structure of the SPAD prediction head is shown in Fig. 8.

III. Results

A. Experiment Environment

1) Model Training Environment: The server utilized in this study operates on the Windows Server operating system, with the deep learning model constructed using the PyTorch framework. The configuration of the server employed in the experiment is detailed in Table IV.

2) Parameter Settings: In model training, parameter configuration is crucial in determining the model's performance.

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TABLE IV
SERVER CONFIGURATION

Category	Configuration
CPU	Intel i7-6800K
GPU	GeForce GTX1080Ti 11GB
System Environment	Windows 10
Framework	Pytorch 1.11.0
Programming voice	Python 3.8

TABLE V
HYPERPARAMETER SETTING

Hyper-parameter	Value
Initial learning rate	0.01
Weight decay	0.0005
Input	640×640px
Max Epoch	150
Batch size	8

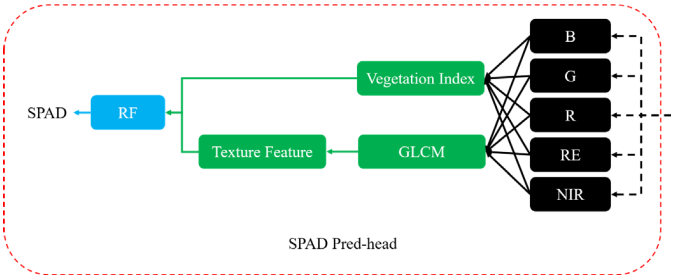


Fig. 8. Structure of SPAD prediction head.

In this experiment, 20% of the multispectral image dataset was used as a validation set, and different parameter combinations were tested by controlling variables. The parameter combination suitable for this experiment was determined through testing. The hyperparameters set in this experiment are shown in Table V. In this particular experiment, the input image was standardized to a resolution of 640 × 640 pixels. The batch size was set at 8, and various spectral channel combinations were employed for training over 150 iterations. The initial learning rate for the training process was set at 0.01, with a weight decay factor of 0.0005.

3) Evaluation Metrics: The performance of the tea tree canopy segmentation model in recognizing the target frame is evaluated using prediction precision (P) and recall rate (R). These metrics are computed according to the following equations:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{10}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{11}$$

In this context, TP, FP, and FN represent the number of correctly classified canopies (true positives), the number of incorrectly classified canopies (false positives), and the number of canopies that the segmentation model failed to classify (false negatives), respectively.

Intersection over union (IoU) serves as a metric to ascertain the accuracy of segmentation. If the IoU, calculated between the pixels predicted by the canopy segmentation and the correctly classified pixels, exceeds 0.5, the pixels are deemed to be correctly classified. The precision and recall rate of the model are calculated by the IoU algorithm. These values, P and R, are then used to calculate the average precision (AP) and mean AP (mAP) according to (12). This process facilitates the evaluation of the model's performance in terms of its segmentation capabilities and its proficiency in detecting and segmenting the tea tree canopy

$$\mathrm{AP} = \int_0^1 P(R)\, dR. \tag{12}$$

In order to evaluate the final SPAD prediction performance of the model, the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE) were used. These metrics, R^2, RMSE, and MAE, are used to compare the model's predicted values with the actual measured values. The corresponding formulas are as follows:

$$R^2 = \frac{\sum_{t=1}^{n} (y - y_m)^2 \left(y_p - y_{pm}\right)^2}{n \sum_{t=1}^{n} (y - y_m)^2 \sum_{t=1}^{n} \left(y_p - y_{pm}\right)^2} \tag{13}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{t=1}^{n} \left(y_p - y\right)^2}{n}} \tag{14}$$

$$\mathrm{MAE} = \frac{\sum_{t=1}^{n} \left|y_p - y\right|}{n}. \tag{15}$$

In the aforementioned formula, y denotes the actual measured value of SPAD, while y_m represents the average of these actual measured SPAD values. y_p stands for the model's predicted SPAD value, and y_{pm} signifies the average value of these predicted SPAD values.

R^2 indicates the proportion of the variability of the dependent variable that the model can explain. Its value range is [0, 1]. The closer it is to 1, the better the model fits the data. RMSE and MAE represent the error between the predicted value and the true value. In theory, RMSE emphasizes a larger error penalty, and MAE represents the robustness of the model.

B. Multispectral Characteristics of Tea Tree Canopy

The ROI for various objects is extracted from the original tea tree UAV multispectral image, and their reflectance is calculated. The average reflectance for each distinct target object within the 450–840-nm spectrum is depicted in Fig. 9. Within the 450–650-nm band, the spectral reflectance of the tea tree canopy falls between that of miscellaneous trees, weeds, branches, drainage channels, and soil. In the 650-nm band, the spectral reflectance of green plants shows a brief decline, while that of other targets increases. In the 730–840-nm band, the spectral reflectance of the tea tree canopy is significantly higher than that of other targets. Given the range of target objects present in the collected spectral images, the multispectral reflectance curve of the tea tree canopy differs from other target objects and exhibits unique reflectance characteristics. Under the 730- and 840-nm wavelength spectrum, the reflectance of the tea tree canopy

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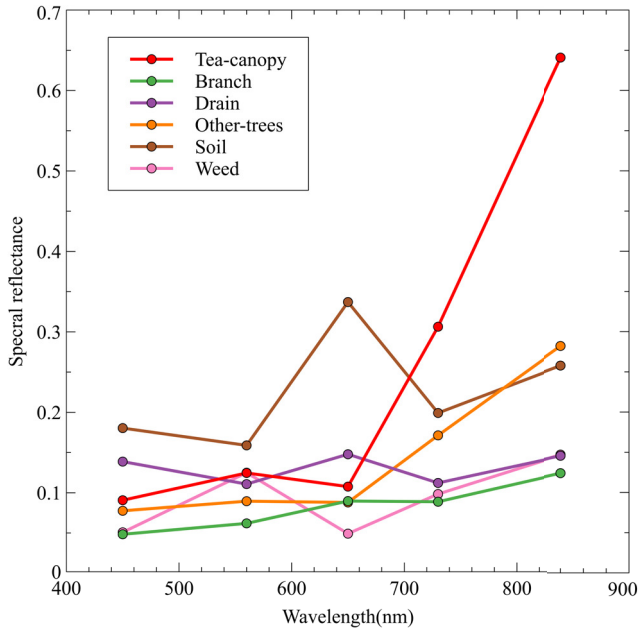


Fig. 9. Average reflectance curves of various targets extracted from multispectral images.

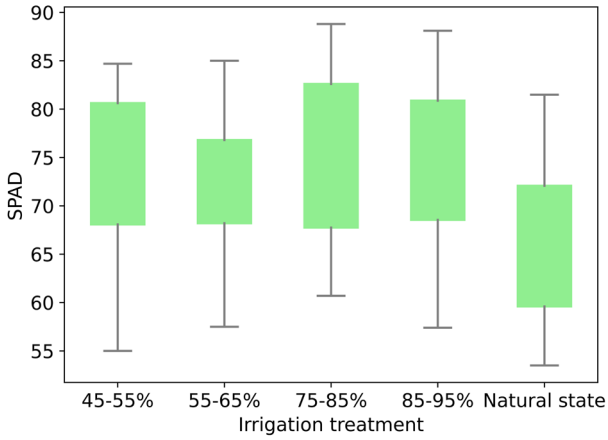


Fig. 10. Distribution of SAPD values under different treatments.

TABLE VI
DESCRIPTIVE STATISTICS OF SPAD DATA UNDER DIFFERENT TREATMENTS

Treatment	SPAD			
	Range	Mean	SD	CV%
45-55%	55.0-84.7	73.93	7.64	10.33
55-65%	57.5-85.0	71.87	7.17	9.97
75-85%	60.7-88.8	76.12	8.70	11.43
85-95%	57.4-88.1	74.48	7.90	10.60
Nature state	53.5-81.5	66.44	8.32	12.53

TABLE VII
TEA TREE CANOPY SEGMENTATION MODEL TEST RESULTS

Model	P(%)	R(%)	mIoU (%)	mAP_{50} (%)	Model size(MB)
DeepLabv3+	89.72	89.90	87.36	-	503.5
DeepLabv3+SF	91.24	94.09	86.14	-	-
YOLOv5	85.50	85.00	88.89	88.96	177.2
YOLOv5+SF	94.77	95.51	90.65	91.90	-
YOLOv8	96.77	99.72	88.90	89.81	137.3
YOLOv8+SF	94.78	96.48	91.61	99.50	-

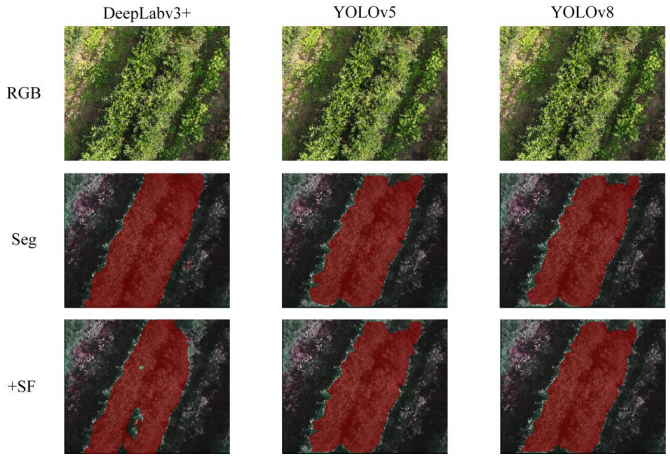


Fig. 11. Partial tea tree canopy multispectral segmentation results map.

is markedly different from that of other objects and exhibits good separability. This indicates that these two wavelengths can provide effective spectral information for the identification of the tea tree canopy.

C. Variation of SPAD in Tea Tree Canopy

A box plot was drawn for the collected SPAD data, as shown in Fig. 10. It was observed that the SPAD of the tea canopy in the natural state was low, while the SPAD of the tea canopy maintained at a certain moisture treatment was high. The mean, SD, and coefficient of variation (CV) were analyzed to further examine the data. Table VI shows the results of the analysis, indicating that the SPAD data for the five moisture treatments ranged from 53.5 to 88.8, with a CV of 11.74%. Analysis of the data under different treatments in the dataset showed that there was a clear difference in the SPAD of the tea canopy after moisture treatment. Therefore, this experiment provided a suitable dataset with significant variability.

D. Tea Tree Canopy Segmentation Results

This study used multispectral images from the UAV tea tree multispectral image and SPAD dataset as the test set for the tea tree canopy segmentation model, and evaluated the detection performance of the YOLOv8 model on tea tree multispectral images. The test results, which are presented in Table VII, indicate that the improved model proposed in this article achieved an R of 99.72% and a P of 96.77% for the tea tree canopy.

A portion of the segmentation result image is displayed in Fig. 11. The spectral image comprises soil blocks, stems, branches, and other objects that are prone to misclassification as areas of the tea tree canopy. Fig. 11 also shows the comparison of the segmentation effects of different models

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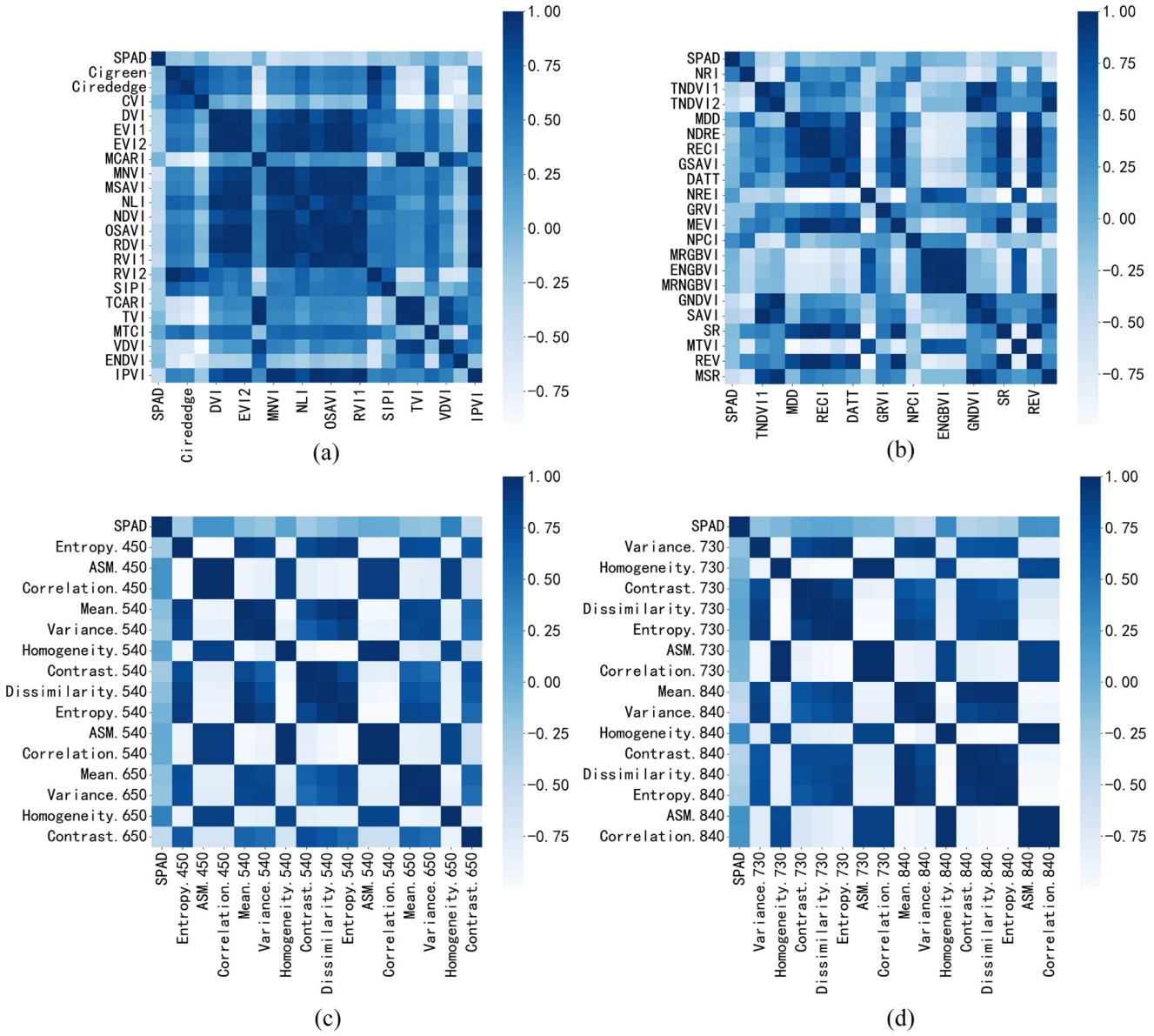


Fig. 12. Correlation analysis diagram. (a) and (b) Correlation between SPAD and vegetation index. (c) and (d) Correlation between SPAD and texture features.

on the tea tree canopy. The DeepLabv3+ model exhibits unexpected positive and negative examples in the canopy segmentation portion of the multispectral image, resulting in subpar performance. While the YOLOv5 model is capable of performing regional segmentation of the tea tree canopy, its classification effect is less than optimal. In contrast, the YOLOv8 model demonstrates effective segmentation on UAV multispectral datasets. Upon integrating the SF module into each model, it is evident that the segmentation effect on the tea tree canopy has significantly improved, rendering it suitable for use as the ROI for feature mapping.

E. Selection of Mapping Features

Upon extracting the ROI for the tea tree canopy from the UAV multispectral image and computing the vegetation indexes and texture features, a Pearson correlation coefficient analysis was conducted between these values and the SPAD. Fig. 12 shows a heat map representing the Pearson correlation between the vegetation indexes, texture features, and SPAD values. The varying colors of the blocks within the figure correspond to different sizes of the correlation coefficient. The darker the color, the stronger the positive correlation between the corresponding two features (horizontal axis and vertical axis). Conversely, the lighter the color, the stronger the negative correlation. As depicted in Fig. 12(a) and (b), most vegetation indexes exhibit a negative correlation with the SPAD values of the tea tree. In Fig. 12(c) and (d), the spectral texture features of the tea trees canopy at wavelengths of 650 and 840 nm demonstrate a strong correlation with the SPAD value, with the majority showing a negative correlation.

For vegetation indexes, select variables with absolute values of correlation coefficients greater than 0.3 These include DVI

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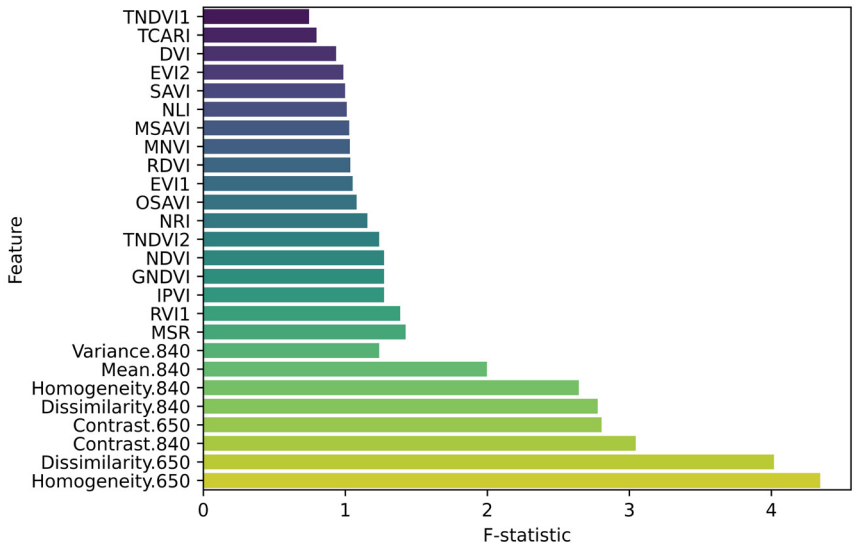


Fig. 13. F-statistic test results of vegetation index and texture characteristics.

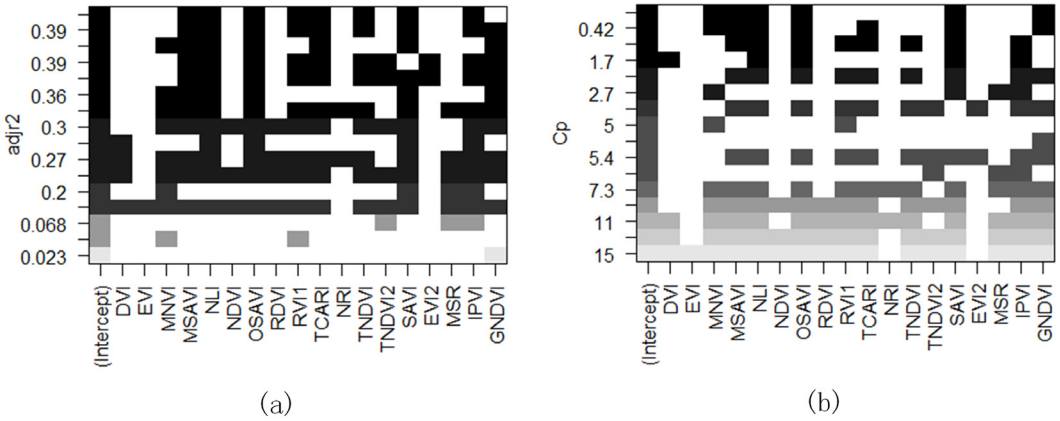


Fig. 14. Vegetation index variable filtering. (a) Adjusted R^2 as an indicator. (b) Mallows' Cp statistic as an indicator.

(–0.334), EVI1 (–0.374), MNVI (–0.595), MSAVI (–0.461), NLI (–0.392), NDVI (–0.474), OSAVI (–0.420), RDVI (–0.399), RVI1 (–0.472), TCARI (–0.329), IPVI (–0.474), NRI (0.435), TNDVI1 (–0.340), TNDVI2 (–0.479), GNDVI (–0.474), SAVI (–0.365), EVI2 (–0.369), and MSR (–0.528), totaling 18 variables. Similarly, for texture features, select variables with an absolute value of the correlation coefficient greater than 0.3. These include homogeneity (0.381), contrast (–0.479), and dissimilarity (–0.439) at 650 nm, as well as the mean (–0.419), variance (–0.506), homogeneity (0.322), contrast (–0.385), and dissimilarity (–0.338) at 840 nm. There are a total of eight variables.

The F-statistic test was performed on these features, and the results are shown in Fig. 13. Among the vegetation indices, MSR (1.424), RVI1 (1.386), and IPVI (1.273) showed the highest F values. In terms of texture features, the F values of homogeneity (4.344), dissimilarity (4.019) at 650 nm, and contrast (3.045) at 840 nm were the highest. Generally speaking, a higher F value means a more significant effect on the SPAD value. The results of the F-statistic test provide a reference for feature screening.

Furthermore, an SWR is performed on the aforementioned variables to construct the optimal variable combination. The SWR method explores different combinations of all available variables, establishes a predictive model for regression analysis, and selects the variable combinations that are effective on the model's interpretive results. The results of the SWR are depicted in Figs. 14 and 15. The left side shows the adjusted R^2 results after feature processing by the SWR method. The right side shows the Mallows' Cp statistics results after feature processing by the SWR method. The black blocks in the figure indicate that the corresponding variables are selected (horizontal axis), and the results of the SWR analysis are sorted from top to bottom (vertical axis).

In the context of vegetation index variable selection, the most effective combination of variables is MSAVI, NLI, OSAVI, RVI1, TCARI, TNDVI1, SAVI, and IPVI, yielding an adjusted R^2 value of 0.39. This combination has been validated as the most efficient set of variables in the predictive model, despite its Mallows' Cp statistic not being the best among all potential combinations. In terms of texture feature variable selection, the optimal combination of variables is homogeneity,

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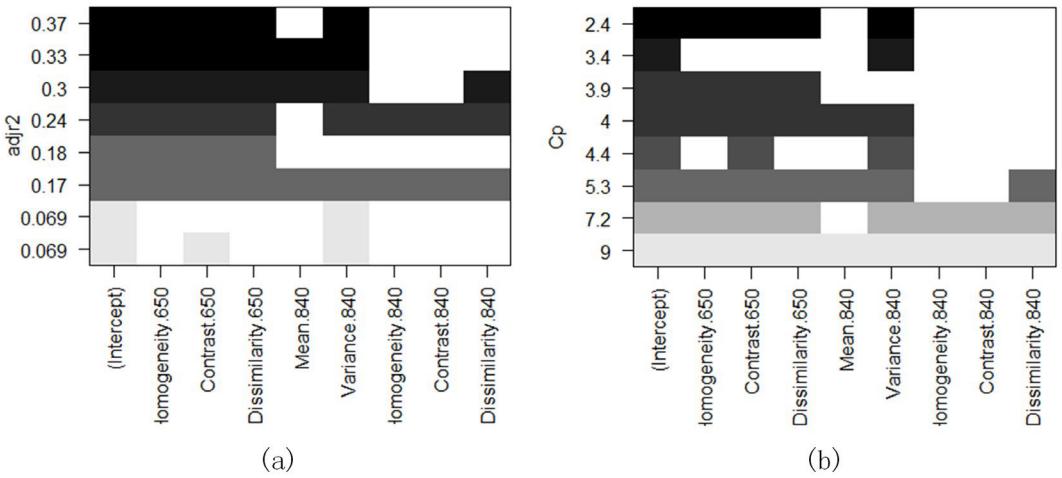


Fig. 15. Texture feature variable filtering. (a) Adjusted R^2 as an indicator. (b) Mallows' Cp statistic as an indicator.

TABLE VIII
MODEL OPTIMAL VARIABLE COMBINATION

Parameter type	Number of parameters	Optimal combination of parameters
Vegetation index	8	MSAVI, NLI, OSAVI, RVI1, TCARI, TNDVI1, SAVI, IPVI
Texture features	4	Homogeneity (650nm), Contrast (650nm), Dissimilarity (650nm), Variance (840nm)
Vegetation index + Texture features	12	MSAVI, NLI, OSAVI, RVI1, TCARI, TNDVI1, SAVI, IPVI, Homogeneity (650nm), Contrast (650nm), Dissimilarity (650nm), Variance (840nm)

TABLE IX
TEST RESULTS OF YOLO-SPAD PREDICTION MODEL

Feature type	Model	Training set			Test set		
		R^2	RMSE	MAE	R^2	RMSE	MAE
Vegetation index	~MLP	0.60	3.54	2.52	0.52	3.89	2.68
	YOLO-SPAD	0.86	2.21	1.86	0.81	2.19	1.74
Texture features	~MLP	0.67	3.24	2.29	0.62	3.46	2.53
	YOLO-SPAD	0.78	2.71	2.22	0.74	2.77	2.21
Vegetation index + Texture features	~MLP	0.67	3.20	2.29	0.63	3.44	2.37
	YOLO-SPAD	0.86	2.12	1.69	0.85	2.14	1.79

contrast, dissimilarity at 650 nm, and variance at 840 nm. The adjusted R^2 for this combination is 0.37, and its Mallows' Cp statistic is at a suitable level.

Drawing from the aforementioned analysis, the final combinations of variables employed to invert the SPAD model for the tea tree canopy are presented in Table VIII.

F. YOLO-SPAD Model Prediction Results

After removing outliers, try to use the above three sets of variables to generate the mapping transformation pairs. These pairs map the image mask to the vegetation indexes and texture features. These mapping transformation pairs are used to establish a SPAD prediction head, and then the fusion model YOLO-SPAD is established through the canopy segmentation pretraining model of tea tree multispectral images to predict and estimate the SPAD value of the tea tree canopy.

The multispectral images and SPAD dataset of the tea tree, captured by the UAV, are divided into training and test sets at a ratio of 7:3. This division is done to test and evaluate the performance of the YOLO-SPAD model. The multispectral image data of the tea tree, captured by the UAV, is used as the input, while the SPAD value of the tea tree canopy serves as the output. The estimation results of the YOLO-SPAD model, obtained using three different sets of variable combinations, are presented in Table IX.

The data presented in Table IX demonstrates the commendable performance of the YOLO-SPAD model, as evidenced by the R^2 surpassing 0.74. This model, YOLO-SPAD, is a product

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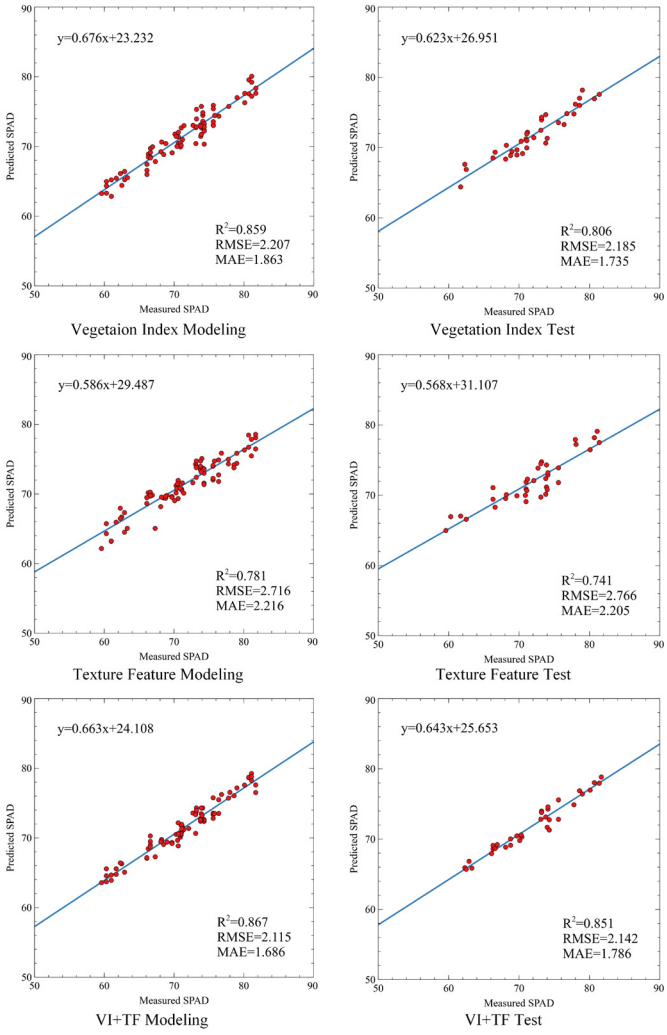


Fig. 16. YOLO-SPAD model prediction results.

of the fusion between the YOLOv8 deep learning algorithm and the RF ensemble learning technique. The robustness of the overall model is noteworthy.

In the estimation of SPAD in tea tree canopies, mapping transformation pairs, which is based on both vegetation indexes and texture features, exhibits a marginally superior performance compared to the YOLO-SPAD model that is established solely on the mapping transformation pairs based on vegetation indexes. The R^2 for the final model's prediction on the test set is 0.85. Furthermore, the model achieved an RMSE of 2.14 and an MAE of 1.79.

Fig. 16 illustrates the predictive outcomes of the YOLO-SPAD model under various variable combinations, further underscoring the model's robust stability and effective estimation capabilities.

IV. Discussion

This study first used a drone equipped with a multispectral camera to collect multispectral images of tea trees. Concurrently, the SPAD of the tea tree canopy was collected at ground level. The reflectance of various targets across five bands was extracted from these multispectral images, serving to validate the efficacy of deep learning methods in identifying the tea tree canopy and performing SPAD inversion. Second, the study trained a tea tree multispectral canopy segmentation model based on YOLOv8 and its improvement, and evaluated its effect and performance. In addition, the prediction model results constructed by the combination of variables composed of different vegetation indexes and texture features were analyzed. Finally, using the best variables as mapping transformation pairs, a SPAD prediction head was designed to construct the YOLO-SAPD model, and the effectiveness of the model was verified. The following discussion pertains to this study.

A. Comparison of Reflectance of Multispectral Images of Target Objects

The challenge of deep learning with multispectral images lies in the abundance of redundant objects and information within the images captured by drones. Furthermore, the spectral images of the same object at different wavelengths present varied representations. The key to achieving more precise segmentation lies in the effective fusion and analysis of multispectral data. Through the reflectance analysis of different target objects via multispectral imaging, it was observed that under the wavelength spectrum of 450, 560, and 650 nm, the reflectance of the tea tree canopy is consistent with each object's reflectance relationship, including branches, miscellaneous trees, weeds, water channels, and soil. The tea tree canopy, miscellaneous trees, and weeds are classified as green plants; hence, their reflectivity between 450 and 650 nm is similar. However, cement ditches and soil exhibit a clear distinction from green vegetation. In the 650-nm band, the reflectivity of green plants shows a brief drop. The exceptional reflectivity of the tea tree canopy at 650 and 840 nm suggests that these two wavelengths may contain more information for extracting the characteristics of the tea tree canopy [72]. This experiment demonstrates that the tea tree canopy exhibits different performances in multispectral images and can achieve results in deep learning image processing to identify and segment the tea tree canopy. Therefore, the feasibility of using deep learning technology in conjunction with UAV spectra to estimate biomass was substantiated.

B. Comparison of Tea Tree Canopy Segmentation Methods Based on Multispectral

The YOLOv8 method was employed to train the annotated tea tree multispectral images using a dataset of 3304 multispectral images. The results indicated that the R of the tea tree canopy is 99.72%, the P is 96.77%, the mIoU is 88.90%, and the mAP is 89.81%. The mIoU values, based on both the YOLOv8 and YOLOv5 models, demonstrated varying degrees of performance enhancement compared to the DeepLabv3+ model for semantic segmentation. This suggests that the instance segmentation model, based on target detection, is effective in multispectral image segmentation. The introduction of the SF module significantly improved the model's segmentation accuracy of the tea tree canopy. In comparison to the unimproved YOLOv8 model, the mIoU of

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the SF-enhanced YOLOv8 model improved by 2.71%. This improvement can be attributed to the SF module's use of multilayer dilated convolution and sampling to fuse multispectral data, thereby further extracting features. This process enhances distinguishability and facilitates the network's ability to synthesize multispectral features to segment the target tea tree canopy.

In terms of model size, the tea tree canopy segmentation model, which is based on the enhanced YOLOv8, is a mere 137.3 MB. This represents a reduction in size by more than 22.5%, rendering the model more lightweight. Consequently, it can be employed for the swift identification and labeling of the ROI in the tea tree canopy. The YOLOv8 model and its improved version demonstrated exceptional segmentation performance on the tea tree canopy. Compared to the traditional method of manually annotating ROI, capturing ROI through the deep learning model is more cost-effective. This provides a solid foundation for the development of UAV-based precision agricultural aviation systems.

C. Analysis of Spectral Characteristics of Tea Tree Canopy

In this study, the tea tree canopy ROI segmented by the tea tree canopy pretraining model was used to calculate different vegetation indexes and texture features. Effective vegetation indexes and textural characteristics play a key role in understanding the health and growth status of tea plants. In order to reduce redundant information and calculation time as much as possible, the correlation between the extracted vegetation index and the texture features and SPAD was analyzed using Pearson correlation. The results revealed a negative correlation between SPAD and the majority of the vegetation indexes and texture features. Among the vegetation indexes, the MNVI exhibited the strongest correlation, with a correlation coefficient of –0.595. Among the texture features, the variance at 840 nm demonstrated the strongest correlation, with a correlation coefficient of –0.506.

Upon further analysis using stepwise linear regression, the optimal combination of vegetation index variables was found to be MSAVI, NLI, OSAVI, RVI1, TCARI, TNDVI1, SAVI, and IPVI. The optimal combination of texture feature variables was homogeneity, contrast, dissimilarity, and variance at 840 nm. These features are associated with the spectral images at 650 and 840 nm, suggesting that these spectral images contain a wealth of information for predicting the SPAD values of tea trees. Therefore, their utilization appears to be more effective.

D. Analysis of SPAD Evaluation Methods for Tea Tree Canopy

In this study, representative multispectral images and SPAD samples were selected to construct tea tree canopy SPAD prediction heads and test the overall model performance. Using the above optimal feature combination, mapping transformation pairs between mask and vegetation indexes and texture features are constructed. This was integrated with the RF model and designed within the YOLOv8 model to propose a tea tree canopy SPAD prediction model named YOLO-SPAD. Upon testing, it was found that in the estimation of tea tree canopy SPAD, mapping transformation pairs based on texture features did not perform as well as the model established based on mapping transformation pairs rooted in the vegetation indexes. This is largely in line with the expectations set by the previous correlation analysis results, indicating that the tea tree canopy is multispectral. The inversion effect of image texture features on SPAD is weaker than that of vegetation indexes. Consequently, the robustness of the final model built is less than the fusion model using vegetation indexes mapping transformation pairs. YOLO-SPAD, which employs vegetation index and texture feature mapping, has shown the best performance in predicting SPAD from tea tree canopy multispectral images, with an R^2 of 0.85, RMSE of 2.14, and MAE of 1.79, indicating a certain predictive ability for SPAD.

However, this study did not consider the time consumed by model calculations. In actual agricultural production, artificial intelligence models are mostly deployed using embedded devices or low-cost computing devices to reduce the investment of farmers who are not already wealthy. The computing performance of these devices is weak, and the deployment of complex models will greatly increase the time cost. The input–output ratio of agricultural production using efficient computing resources is large. This aspect needs to be incorporated in future research. In addition, increasing the selection of mapping features, further improving the model, and improving the estimation effect of YOLO-SPAD are also future research directions.

V. Conclusion

In the field of agricultural remote sensing, the canopy area of the target crop is typically designated as the ROI for biomass analysis. However, due to inaccuracies inherent in manual vision, the tasks of identifying and selecting the crop canopy ROI, as well as estimating the SPAD of the tea tree canopy, pose significant challenges. As such, there is a pressing need for a model that can automatically identify the tea tree canopy and predict SPAD. This study was conducted to address this issue.

This study was conducted in a tea garden where tea trees are cultivated in a standardized fashion. A drone, equipped with a multispectral camera, was utilized to capture multispectral images of the tea trees within the garden. Based on the multispectral images of tea tree canopies, the spectral characteristics of the tea tree canopies were analyzed to verify the feasibility of deep learning processing using multispectral images of tea trees. This article proposes a tea tree canopy SPAD prediction model, YOLO-SPAD, based on UAV multispectral images. This is a new method to estimate the SPAD of tea leaves using UAV multispectral images, providing a better method for crop biomass monitoring in agricultural production. Through the Pearson correlation coefficient and SWR method, the correlations between SPAD and 43 multispectral vegetation indexes and 40 texture features for SPAD were analyzed. Eight vegetation indixes (MSAVI, NLI, OSAVI, RVI1, TCARI, TNDVI1, SAVI, and IPVI) and four texture features [homogeneity (650 nm), contrast (650 nm), dissimilarity (650 nm), and variance (840 nm)] were selected. Based on these features, the mapping transformation relationship was constructed

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between the segmented ROI and different features. Based on YOLOv8, SF is proposed to more effectively fuse multispectral features, achieving a mIoU of 91.61% and a mAP of 99.50%. This improved model effectively reduces accidental segmentation, and significantly improves the accuracy of segmentation results. A SPAD prediction head was designed and incorporated into the model, enabling the estimation of the SPAD value of the tea tree canopy using UAV multispectral images on representative images. Test results showed an R^2 of 0.85, an RMSE of 2.14, and an MAE of 1.79 for SPAD prediction. This work focused on predicting the SPAD of tea trees, and the proposed method can be extended to predict other crops and additional biomass indicators, providing a concept for the future development of multispectral plant protection UAVs. This model can further collaborate with the UAV host computer software to achieve more precise results, thereby facilitating efficient precision agricultural monitoring. Furthermore, this work opens up several promising avenues for future research. One is to reduce the number of parameters to make the model lightweight. Another is to add mapping features and explore its applicability to other crop biomass indicators.

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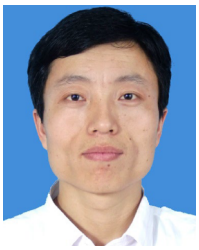
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Jiaxing Xie received the bachelor's degree in applied physics and the master's degree in computer application technology from Harbin University of Science and Technology, Harbin, China, in 2002 and 2005, respectively, and the Doctorate degree in agricultural electrification and automation from South China Agricultural University, Guangzhou, China, in 2016.

He is currently an Associate Professor of agricultural computer and electronic technology at the College of Electronic Engineering (College of Artificial Intelligence), South China Agricultural University. He is committed to the application research of the Internet of Things technology and electronic technology in agriculture, focusing on the acquisition and monitoring of agricultural remote sensing information, including the acquisition, modeling, and control of environmental parameters and phenotypic information in rice fields, orchards, water environments, and other fields.

Liye Chen, photograph and biography not available at the time of publication.

Jiatao Wu, photograph and biography not available at the time of publication.

Zonghong Li, photograph and biography not available at the time of publication.

Yazhong Chen, photograph and biography not available at the time of publication.

Meiyi Lu, photograph and biography not available at the time of publication.

Yingxin Zou, photograph and biography not available at the time of publication.

Peng Gao, photograph and biography not available at the time of publication.

Zheng Shen, photograph and biography not available at the time of publication.

Daozong Sun, photograph and biography not available at the time of publication.

Weixing Wang, photograph and biography not available at the time of publication.



Jun Li received the Ph.D. degree from South China University of Technology, Guangzhou, China, in 2007.

He is the Dean of the College of Engineering and a Professor of agricultural engineering at South China Agricultural University, Guangzhou. His research areas include modern agricultural equipment, modern design methods of agricultural remote sensing, and the technology of intelligent control using it.

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